

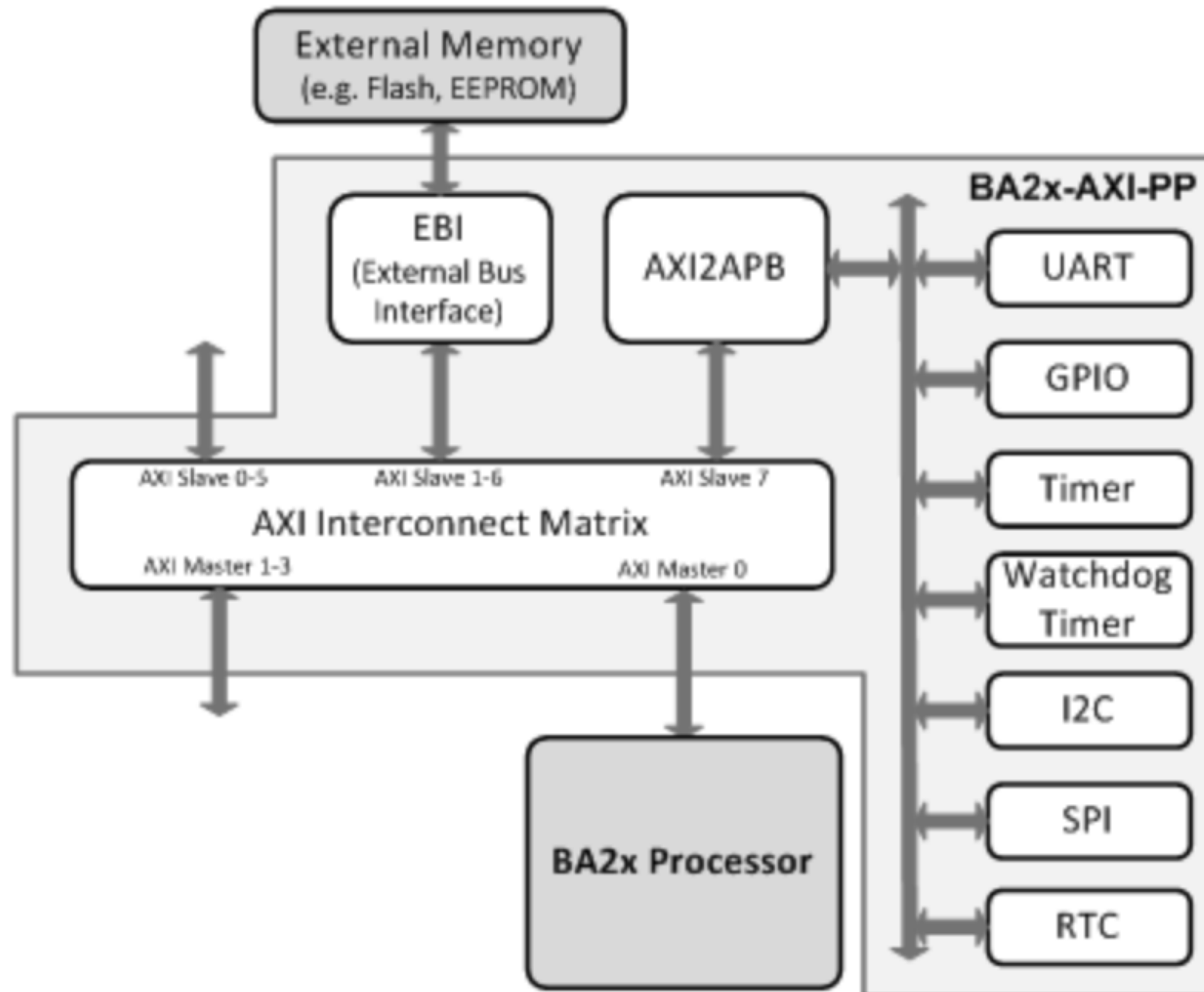


AMBA PROTOCOLS

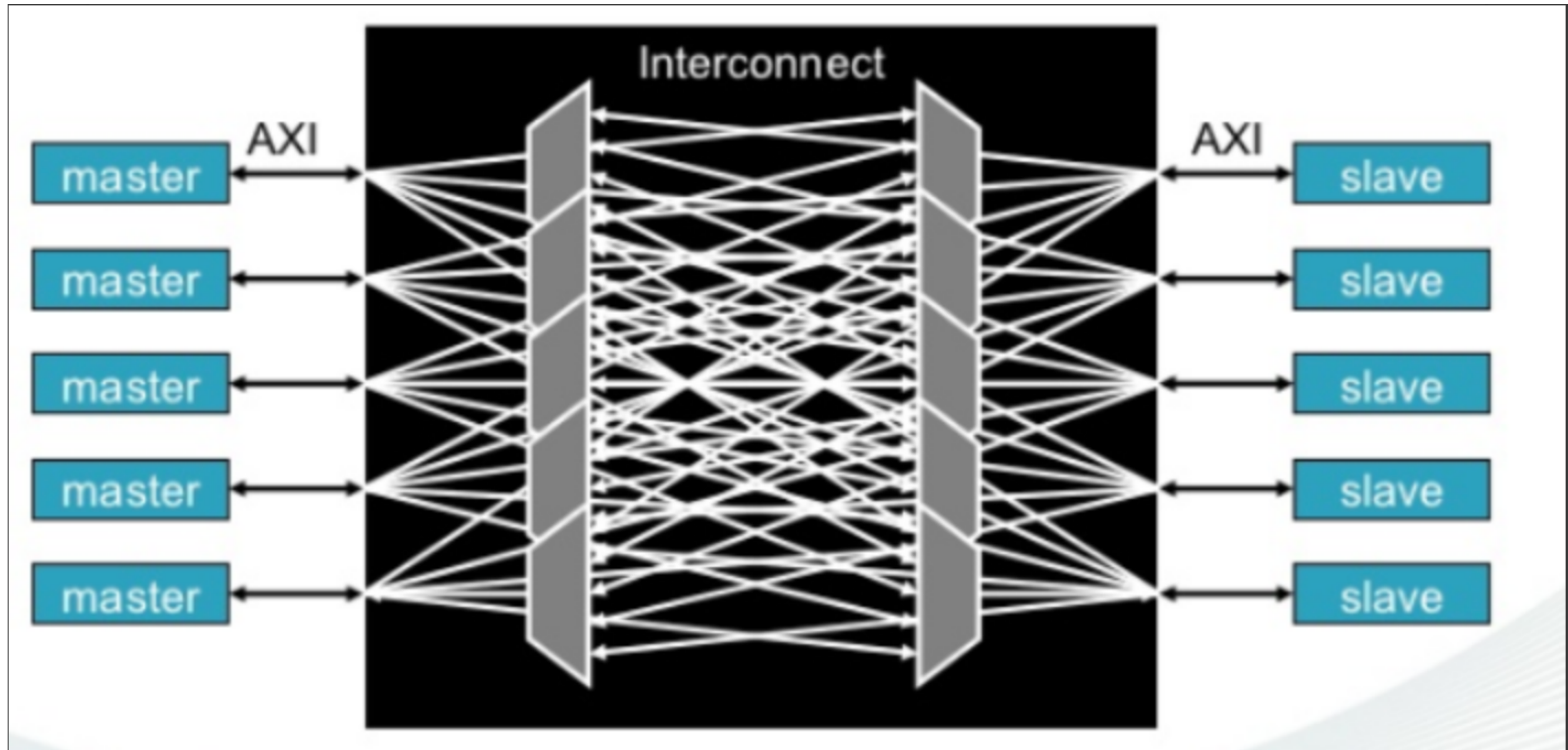
AHB

ADVANCED HIGH PERFORMANCE BUS

AXI INTERCONNECT IN SOC



AXI INTERCONNECT IN NOC



CLASSIFICATION OF AMBA

Key AMBA Specifications

AMBA generation:		AMBA 2	AMBA 3	AMBA 4	AMBA 5
CHI Coherent Hub Interface	CHI is a credited coherency protocol, layered architecture for scalability				CHI
ACE AXI coherency Extensions	ACE is superset of AXI – brings system-wide coherency across multicore clusters			ACE +Lite	ACE5 +Lite
AXI Advanced eXtensible Interface	AXI supports separate A/D phases, bursts, multiple outstanding addresses, OoO responses		AXI3	AXI4 +Lite, +Stream	AXI5
AHB Adv. High-performance Bus	AHB supports 64/128 bit, multi-master. AHB-Lite for single masters	AHB	AHB +Lite		AHB5 +Lite
APB Advanced Peripheral Bus	System bus for low b/w peripherals	APB2	APB3	APB4	

AMBA SPECIFICATION (FIRST VERSION) DEFINES TWO BUSES/INTERFACES:

- Advanced System Bus (ASB)
- Advanced Peripheral Bus (APB)

AMBA 2 SPECIFICATION DEFINES THREE BUSES/INTERFACES:

- Advanced High-performance Bus (AHB) - widely used on ARM7, ARM9 and Arm Cortex-M based designs
- Advanced System Bus (ASB)
- Advanced Peripheral Bus (APB2 or APB)

AMBA 3 SPECIFICATION DEFINES FOUR BUSES/INTERFACES:

- Advanced Extensible Interface (AXI3 or AXI v1.0) - widely used on Arm Cortex-A processors including Cortex-A9
- Advanced High-performance Bus Lite (AHB-Lite v1.0)
- Advanced Peripheral Bus (APB3 v1.0)
- Advanced Trace Bus (ATB v1.0)

THE AMBA 4 SPECIFICATION DEFINES FOLLOWING BUSES/INTERFACES:

- AXI Coherency Extensions (ACE) - widely used on the latest Arm Cortex-A processors including Cortex-A7 and Cortex-A15
- AXI Coherency Extensions Lite (ACE-Lite)
- Advanced Extensible Interface 4 (AXI4)
- Advanced Extensible Interface 4 Lite (AXI4-Lite)
- Advanced Extensible Interface 4 Stream (AXI4-Stream v1.0)
- Advanced Trace Bus (ATB v1.1)
- Advanced Peripheral Bus (APB4 v2.0)
- AMBA Low Power Interfaces (Q-Channel and P-Channel)

THE AMBA 5 SPECIFICATION DEFINES THE FOLLOWING BUSES/INTERFACES:

- AXI5, AXI5-Lite and ACE5 Protocol Specification
- Advanced High-performance Bus (AHB5, AHB-Lite)
- Coherent Hub Interface (CHI) [3]
- Distributed Translation Interface (DTI)
- Generic Flash Bus (GFB)

AHB FEATURES

Burst transfers

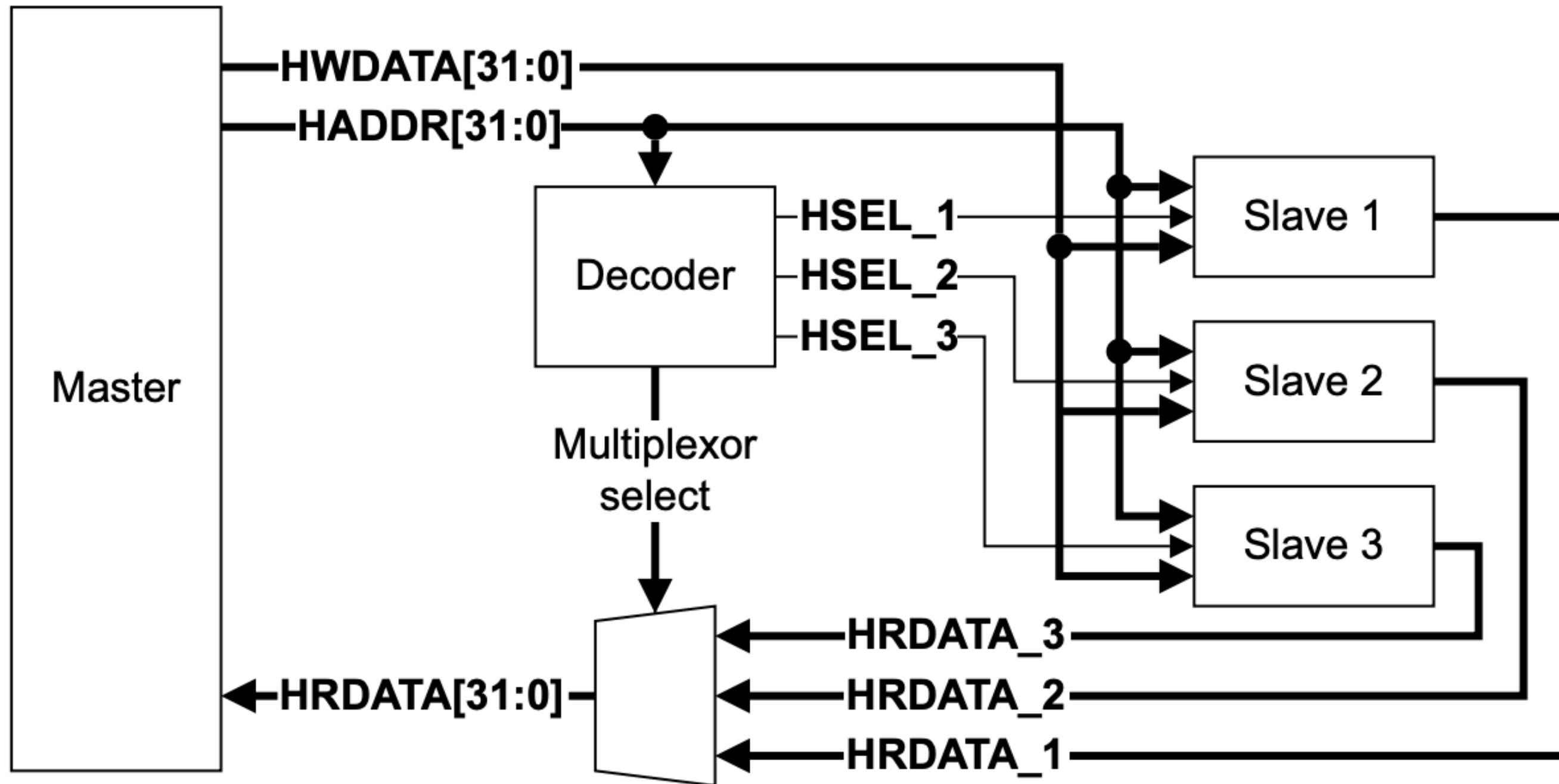
Split Transactions

Wider data bus configurations (64/128 bits)

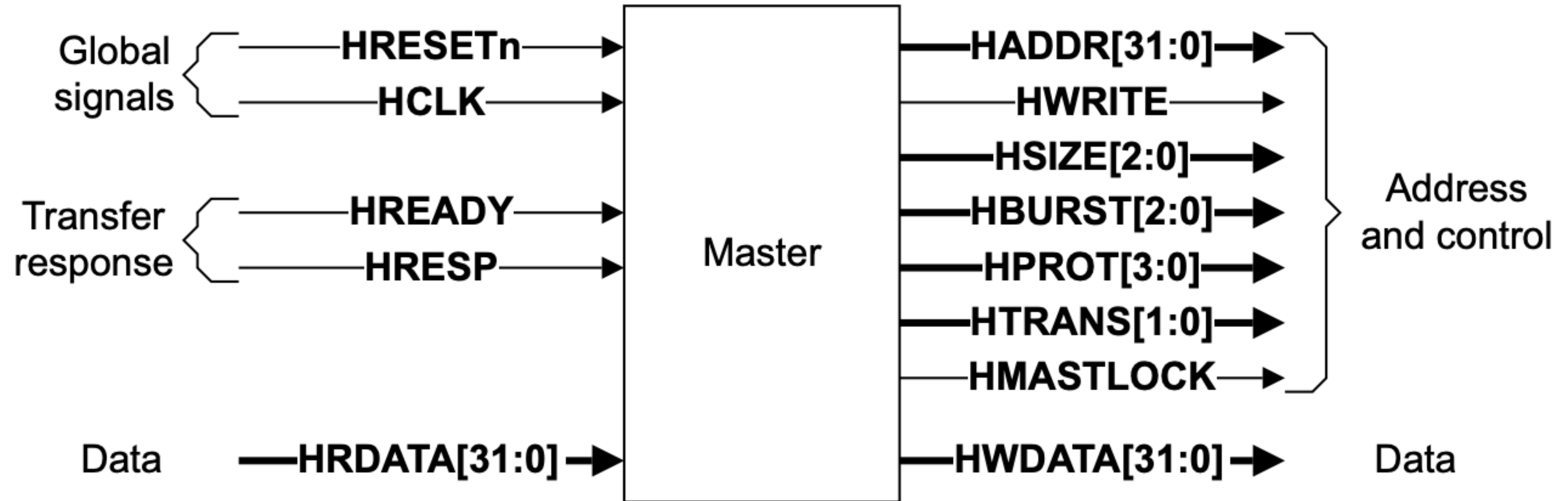
Single-clock edge operation

Single-cycle bus master handover

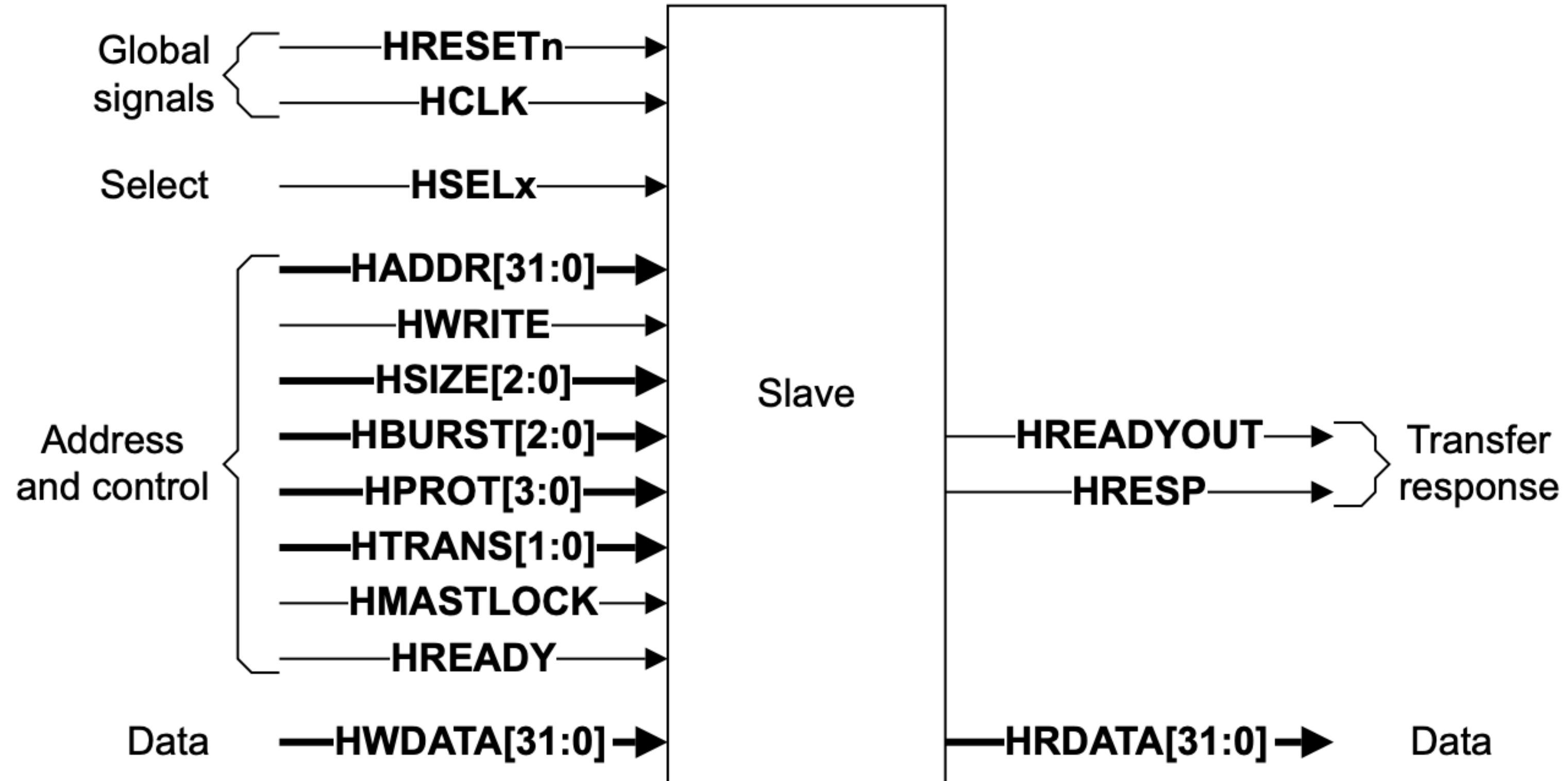
1 AHB MASTER, 3 AHB SLAVE BLOCK DIAGRAM



MASTER INTERFACE



SLAVE INTERFACE



AHB TRANSFERS

- Type of Transfers
- Transfer Size
- Simple Transfers
- Locked Transfers
- Burst operation
- Waited Transfers

TYPE OF TRANSFERS - HTRANS[1:0]

IDLE (00) – Indicate a new transfer

- ☒ No data transfer is required
- ☒ Used when the master is granted the bus without performing transfer
- ☒ The slave must provide a zero wait state OKAY response

BUSY (01)

- ☒ Allow master to insert IDLE cycle in the middle of bursts of transfers
- ☒ Address and control signals must reflect the next transfer in the burst
- ☒ The slave must provide a zero wait state OKAY response

NONSEQ (10) – Indicate a new transfer

- ☒ Indicate the first transfer of a burst or a single transfer
- ☒ Address and control signals are unrelated to previous transfer

SEQ (11)

- ☒ Indicate a remaining transfer in a burst

Address = Address of previous transfer + size in byte

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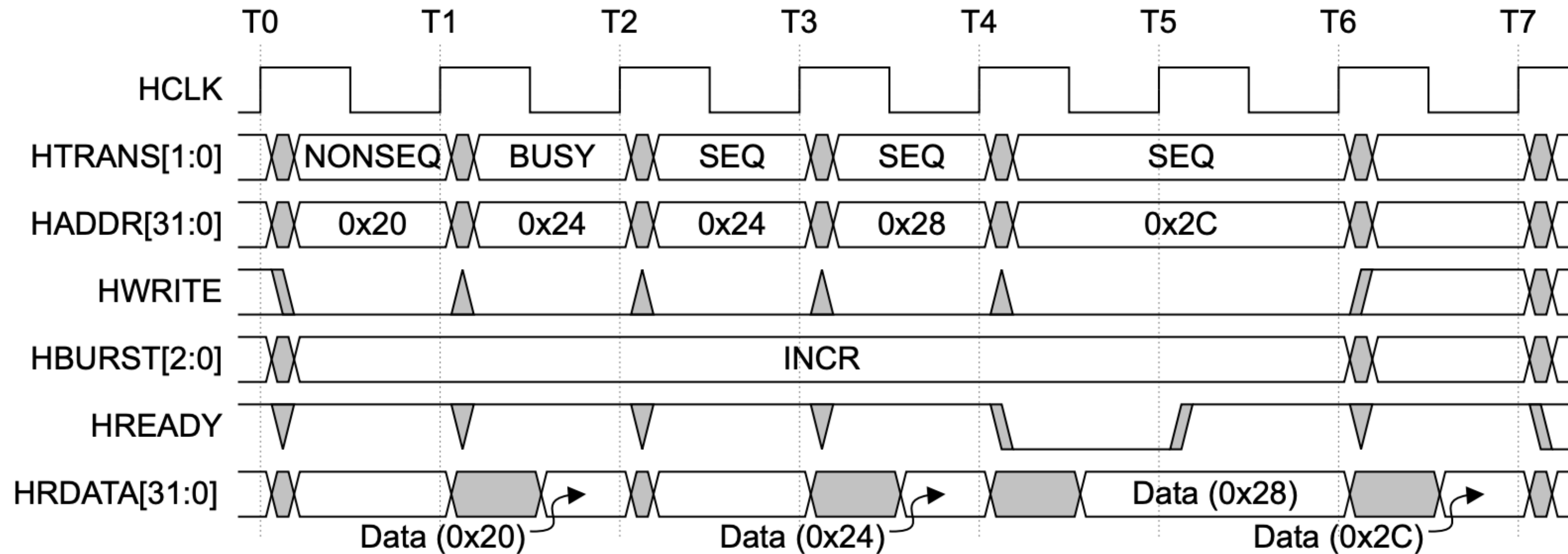
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TYPE OF TRANSFERS - HTRANS[1:0]



TRANSFER SIZE - HSIZE[2:0]

HSIZE[2]	HSIZE[1]	HSIZE[0]	Size (bits)	Description
0	0	0	8	Byte
0	0	1	16	Halfword
0	1	0	32	Word
0	1	1	64	Doubleword
1	0	0	128	4-word line
1	0	1	256	8-word line
1	1	0	512	-
1	1	1	1024	-

SIMPLE TRANSFERS

- **Masters must be granted bus for access before transfer**

- ☒ Masters assert request signals
- ☒ Arbiter indicates when the master will be granted use of the bus
- ☒ An AHB-bus transfer

- **Address phase**

A single cycle in which the master drives the address and control signals

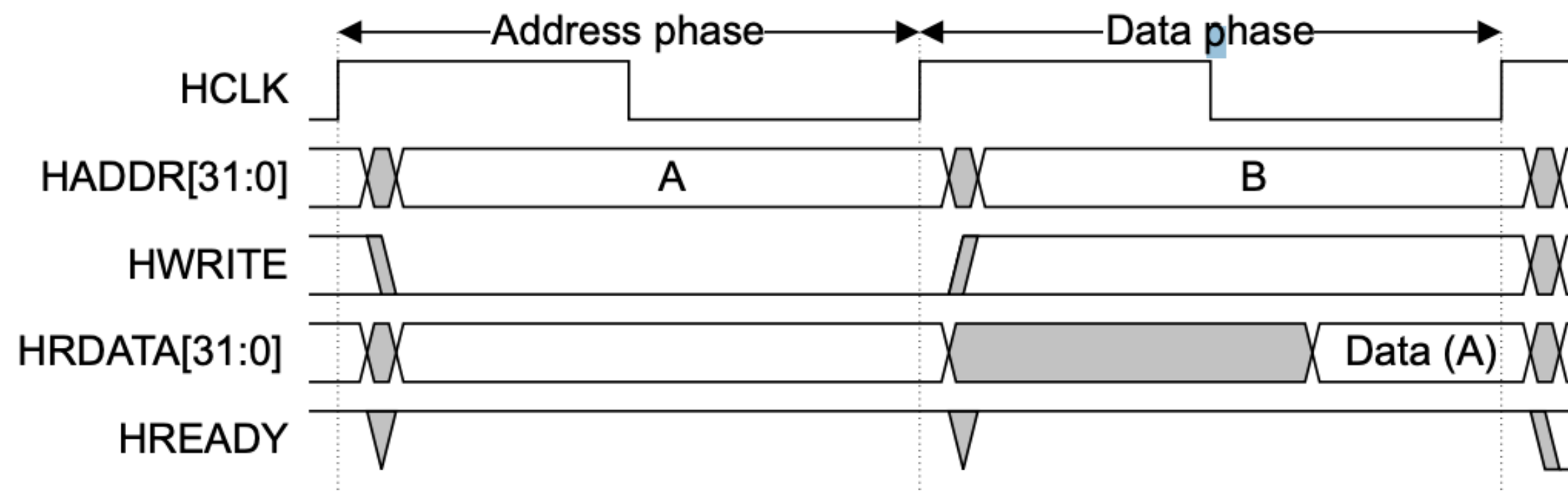
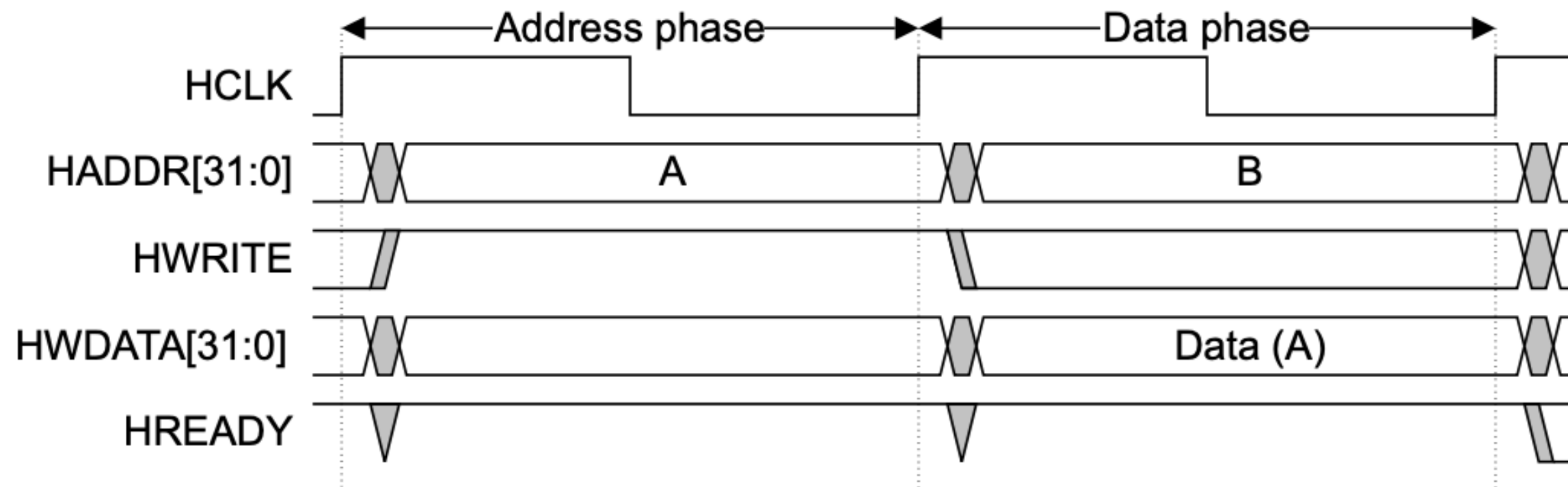
Indicate direction, width of transfer, and if the transfer forms part of a burst

- ☒ Each address correspond to 1-byte data (byte-addressable)

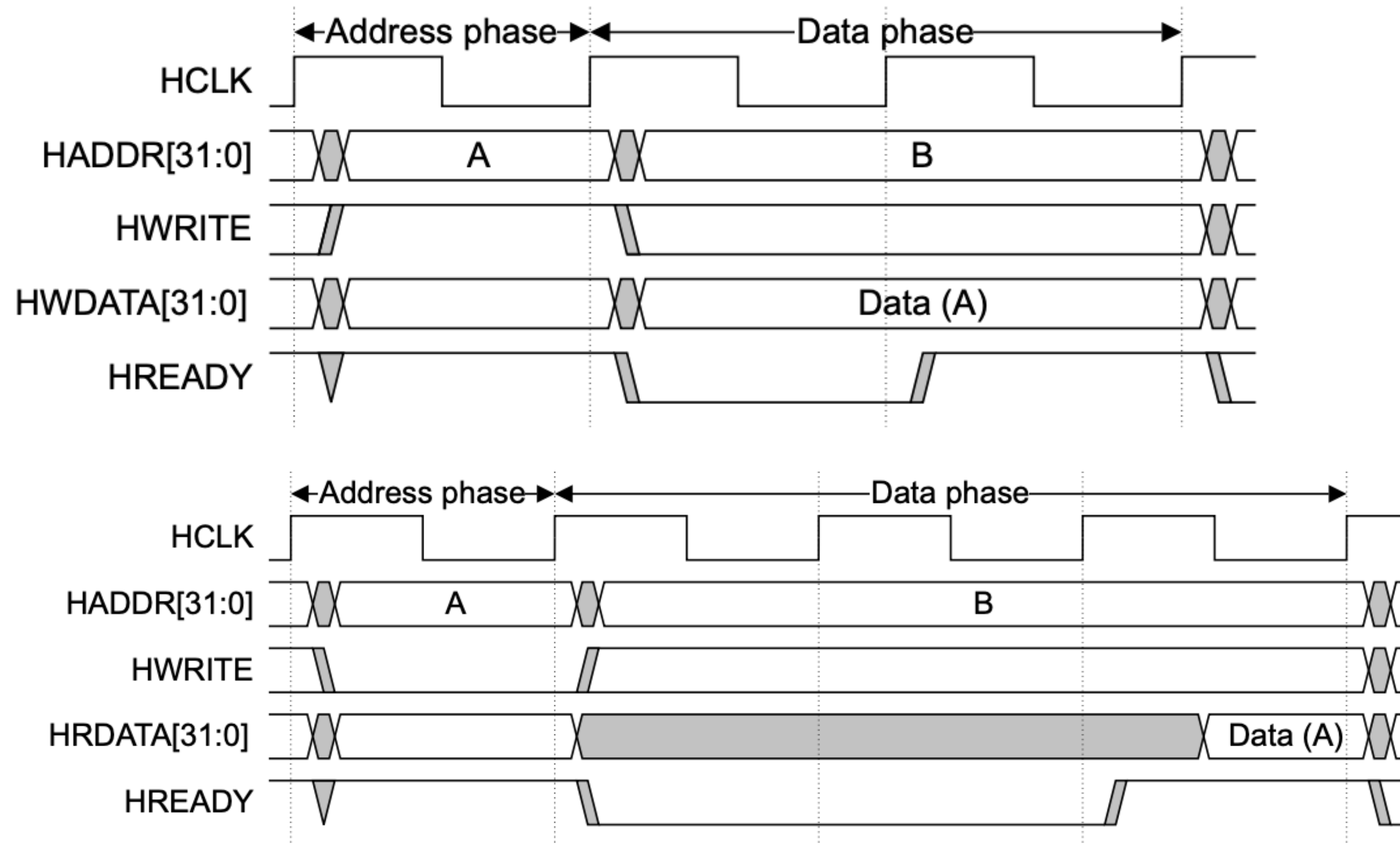
- **Data phase**

- ☒ One or more cycles controlled by HREADY from slave
- ☒ The slaves sample and process the data
- ☒ Address phase and data phase of different transactions are overlapped

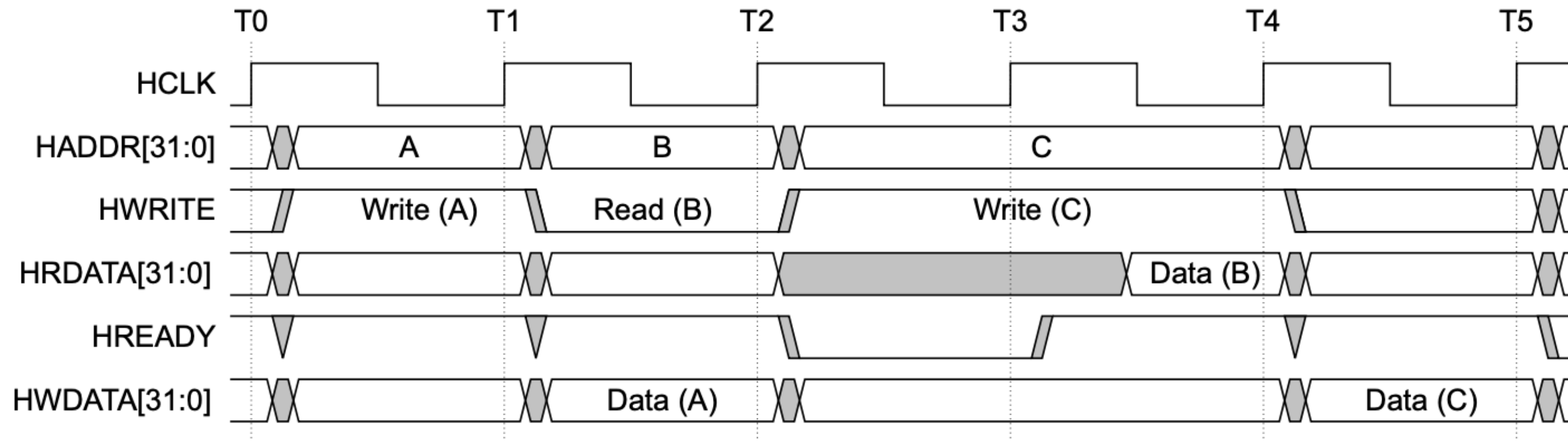
WRITE / READ TRANSFERS



WITH WAIT STATES

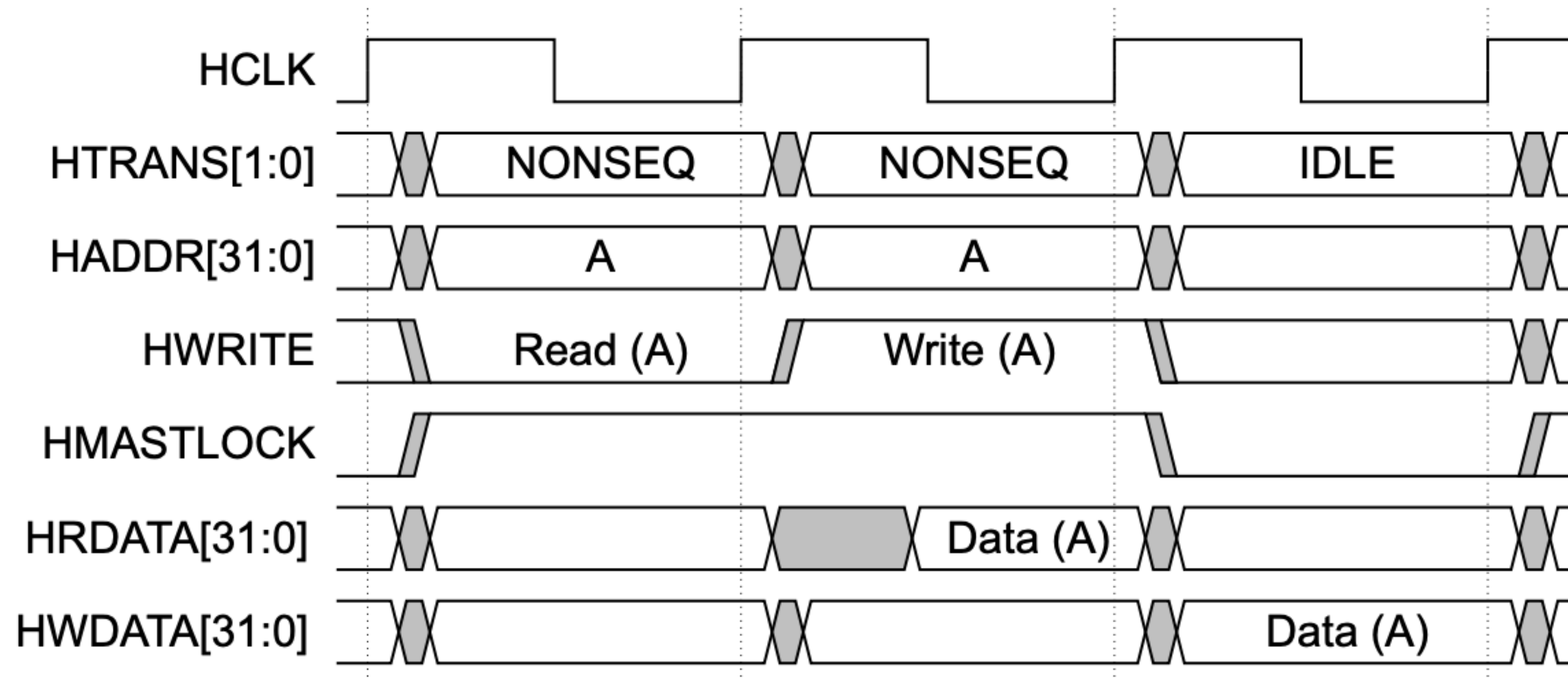


MULTIPLE TRANSFERS



LOCKED TRANSFERS - HMASTLOCK

If the master requires locked accesses then it must also assert the HMASTLOCK signal. This signal indicates to any slave that the current transfer sequence is indivisible and must therefore be processed before any other transfers are processed.



BURST OPERATION

- 4/8/16-beat bursts, as well as undefined-length bursts ☒
- Burst size indicates the number of beats in the burst ☒
- Valid data transferred = (# of beats) x (data size in each beat) ☒ Data size in each beat is indicated in HSIZE[2:0] ☒

Incrementing bursts ☒

Access sequential locations with the address of each transfer in the burst being an increment of the previous address ☒

An incremental burst can be of any length, but the upper limit is set by the fact that address must not cross a 1KB boundary ☒

Wrapping bursts ☒

If the start address of the transfer is not aligned to the total number of bytes in a burst (size x beats) then the address of the transfers in the burst will wrap when the boundary is reached ☒

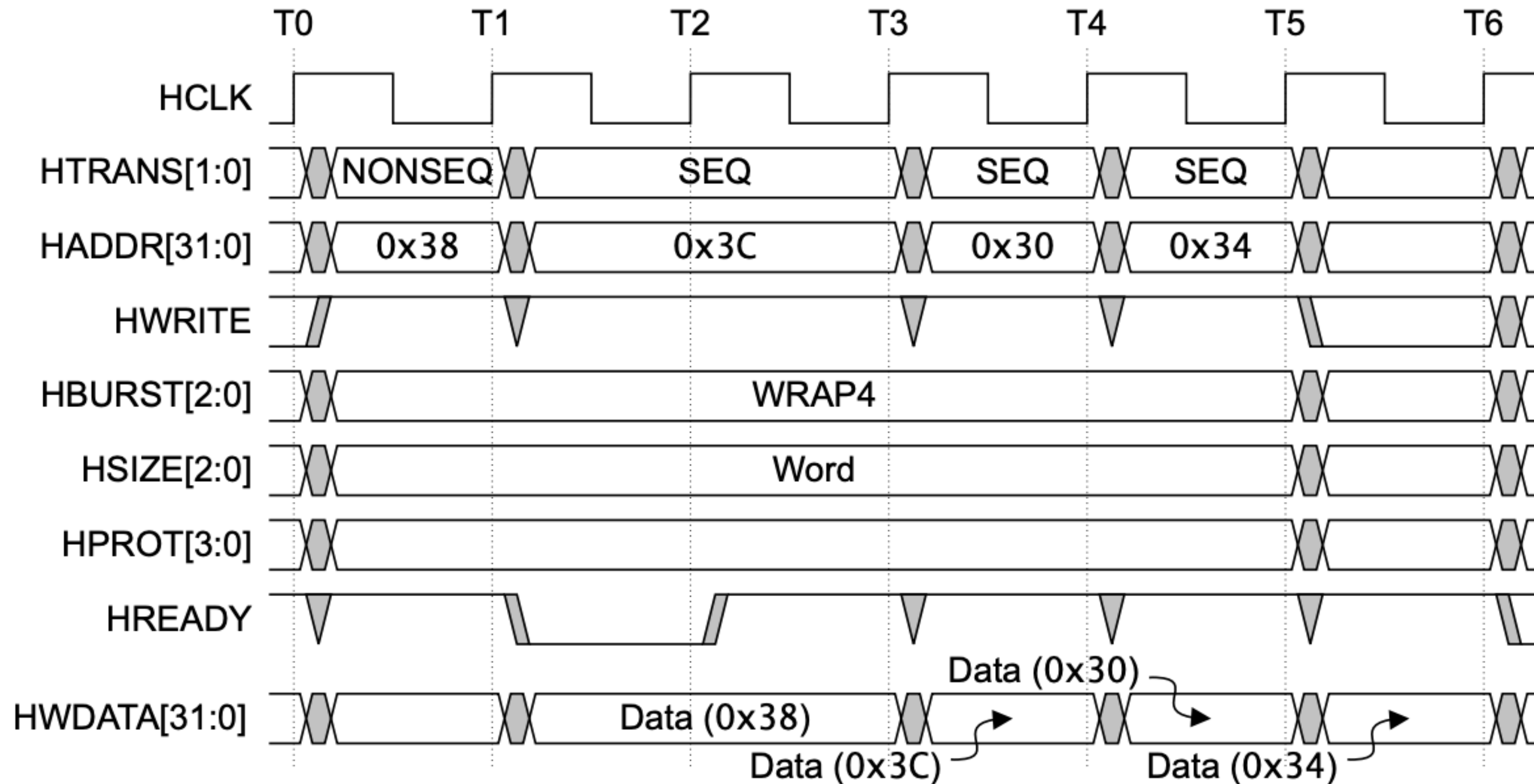
BURST SIGNAL ENCODING

HBURST[2:0]	Type	Description
0b000	SINGLE	Single transfer burst
0b001	INCR	Incrementing burst of undefined length
0b010	WRAP4	4-beat wrapping burst
0b011	INCR4	4-beat incrementing burst
0b100	WRAP8	8-beat wrapping burst
0b101	INCR8	8-beat incrementing burst
0b110	WRAP16	16-beat wrapping burst
0b111	INCR16	16-beat incrementing burst

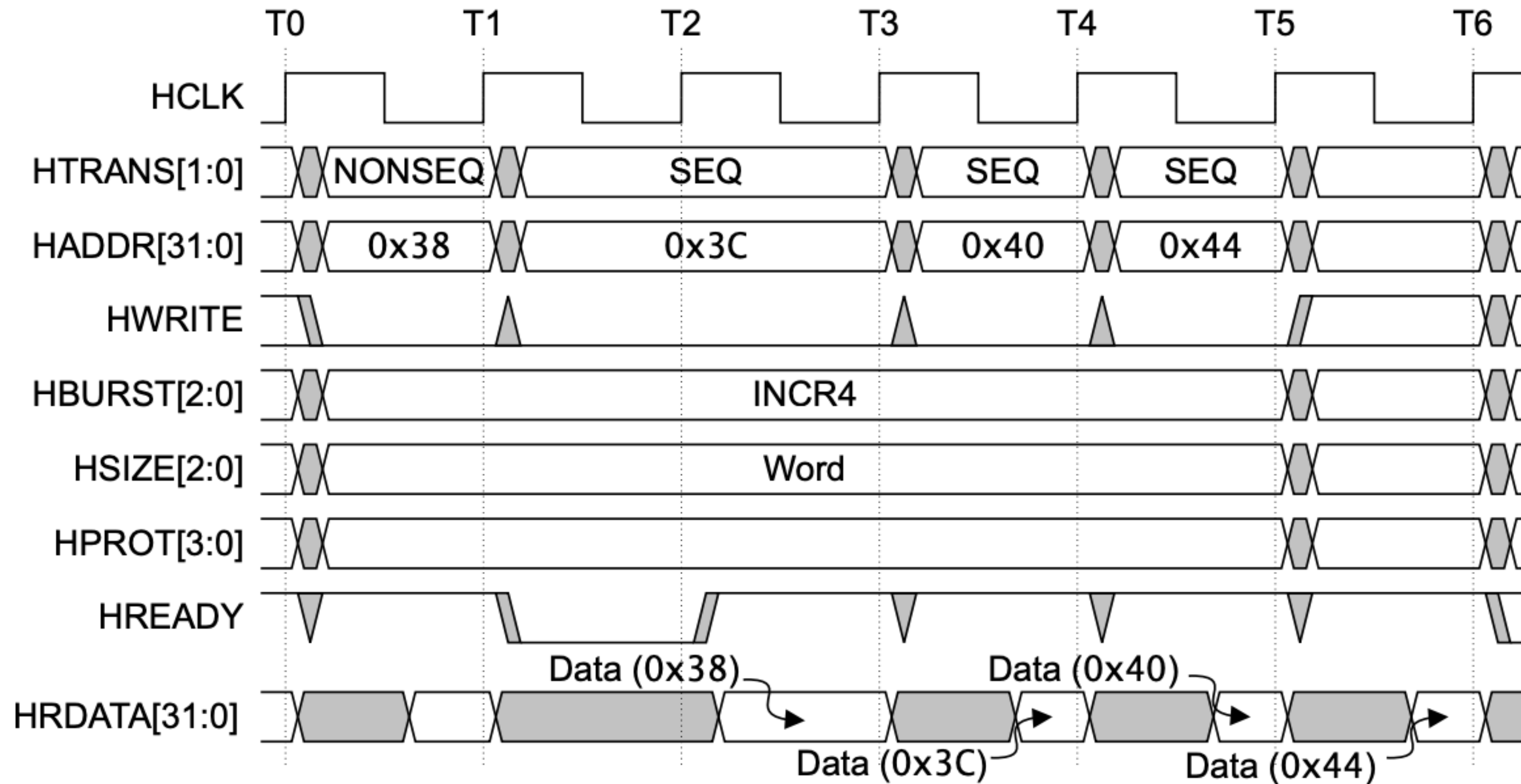
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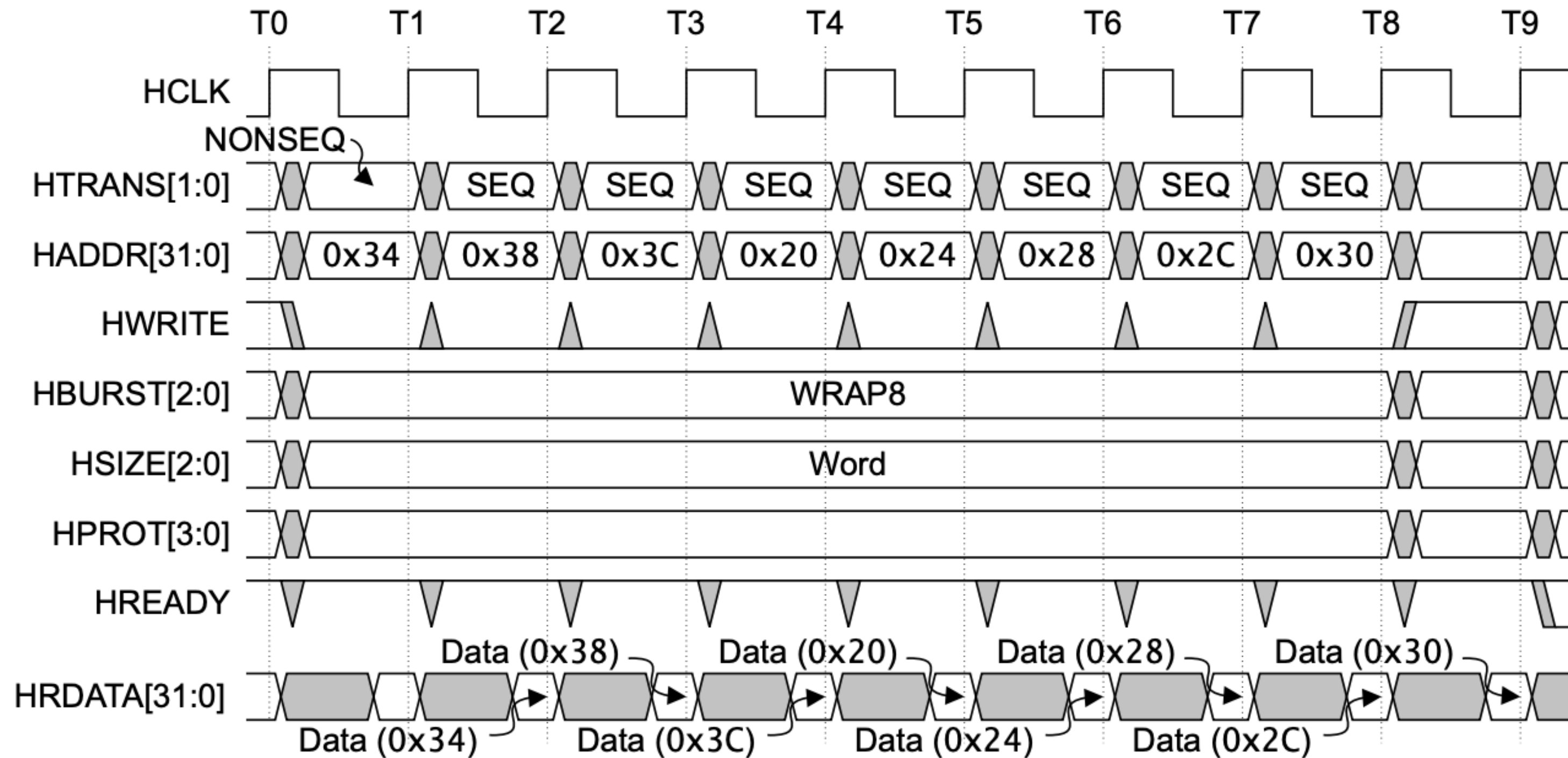
FOUR-BEAT WRAPPING BURST, WRAP4



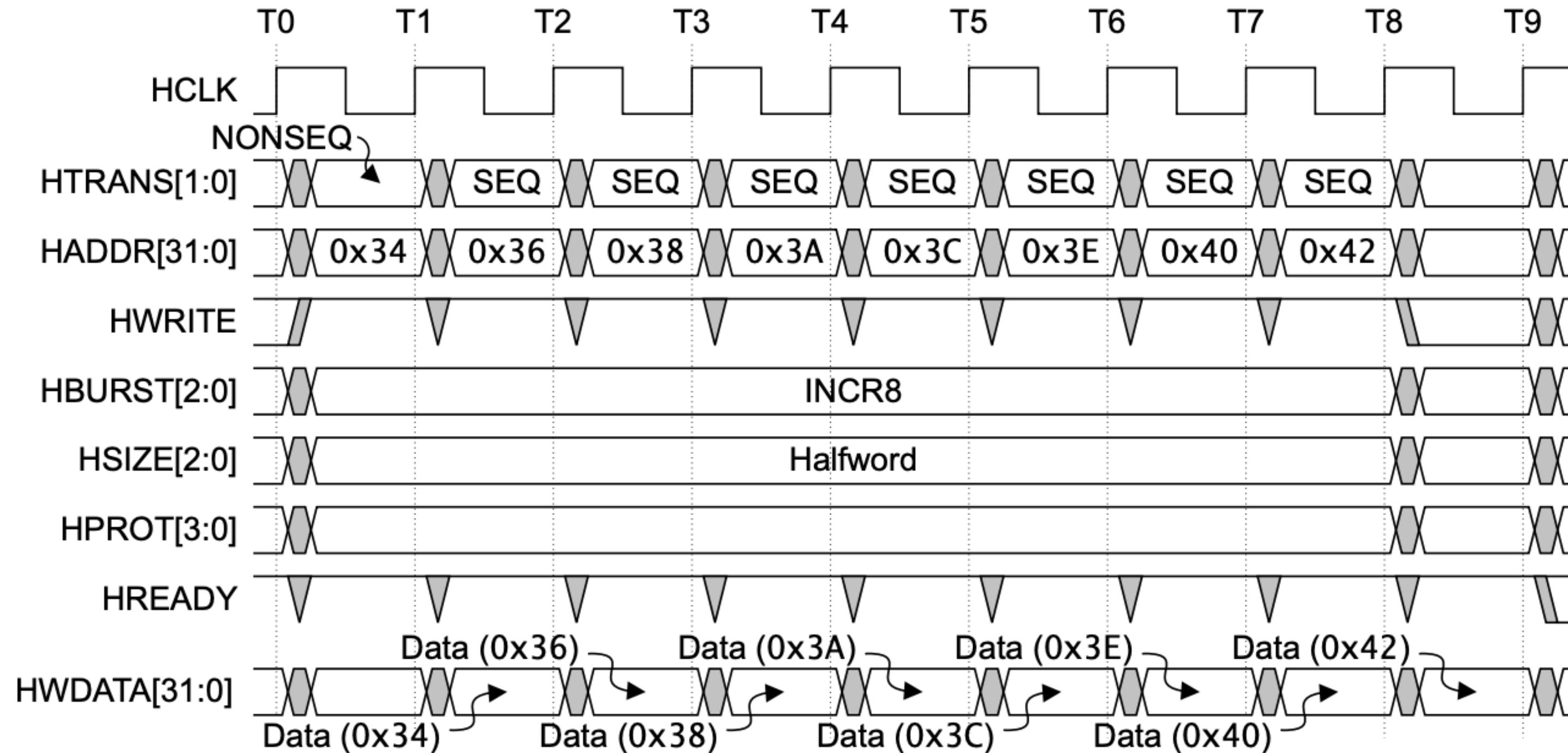
FOUR-BEAT INCREMENTING BURST, INCR4



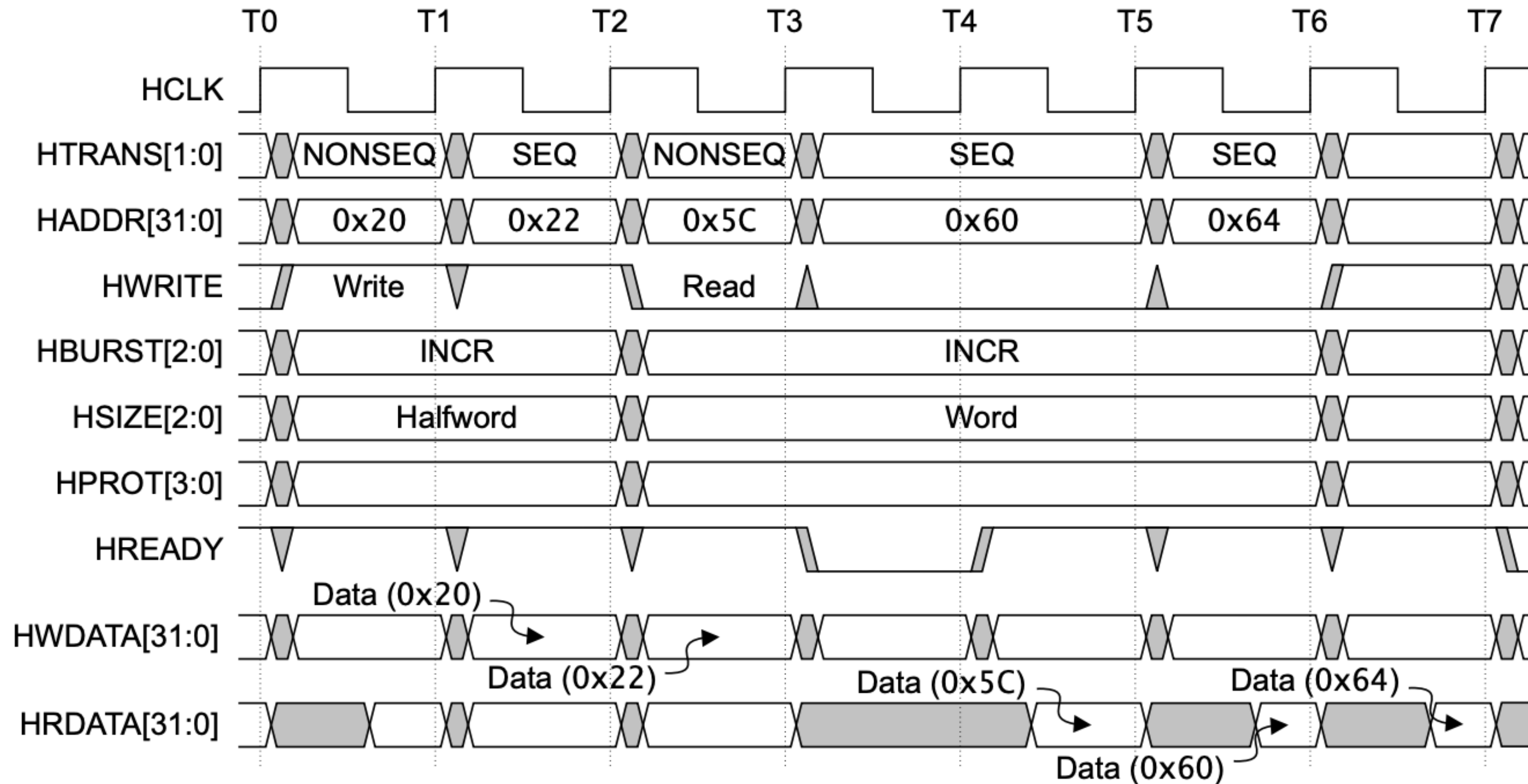
EIGHT-BEAT WRAPPING BURST, WRAP8



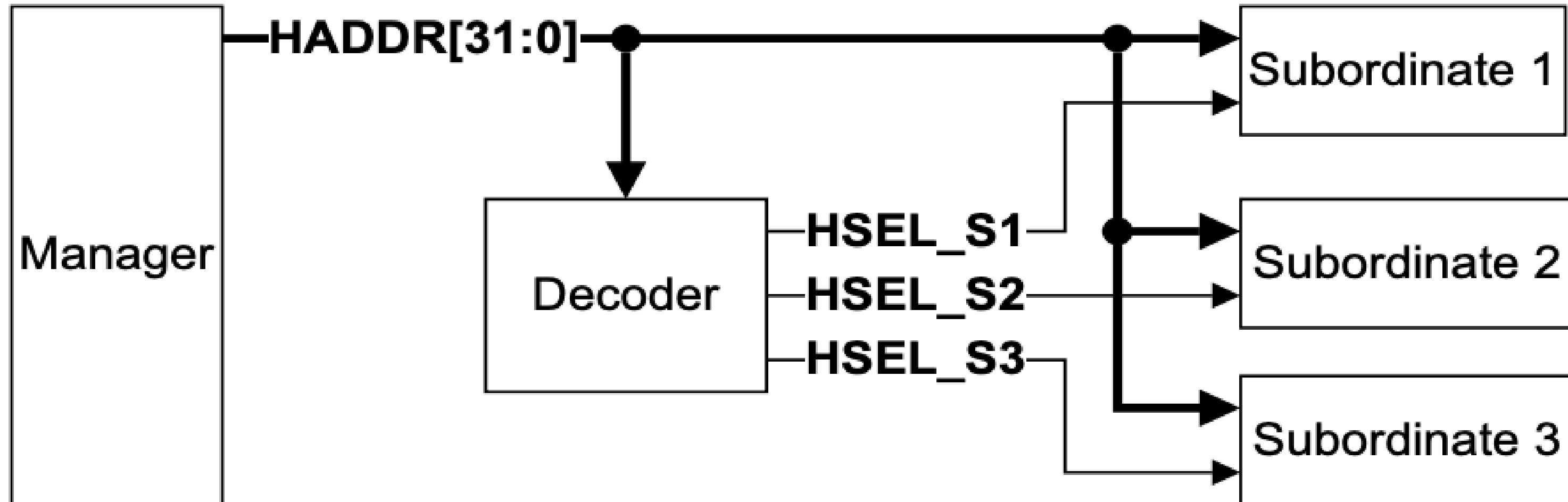
EIGHT-BEAT INCREMENTING BURST, INCR8



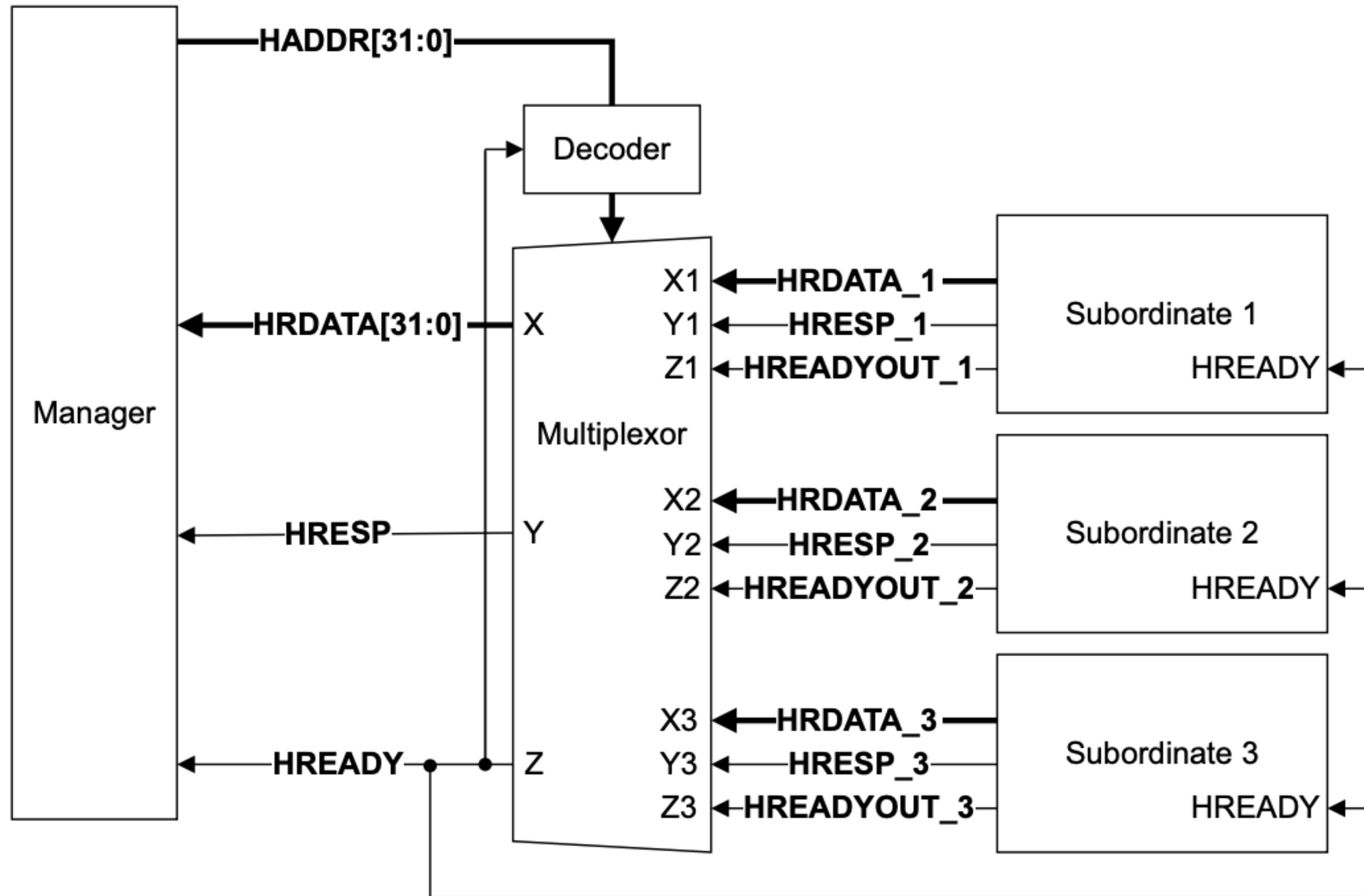
EXAMPLE OF UNDEFINED LENGTH BURST OPERATION



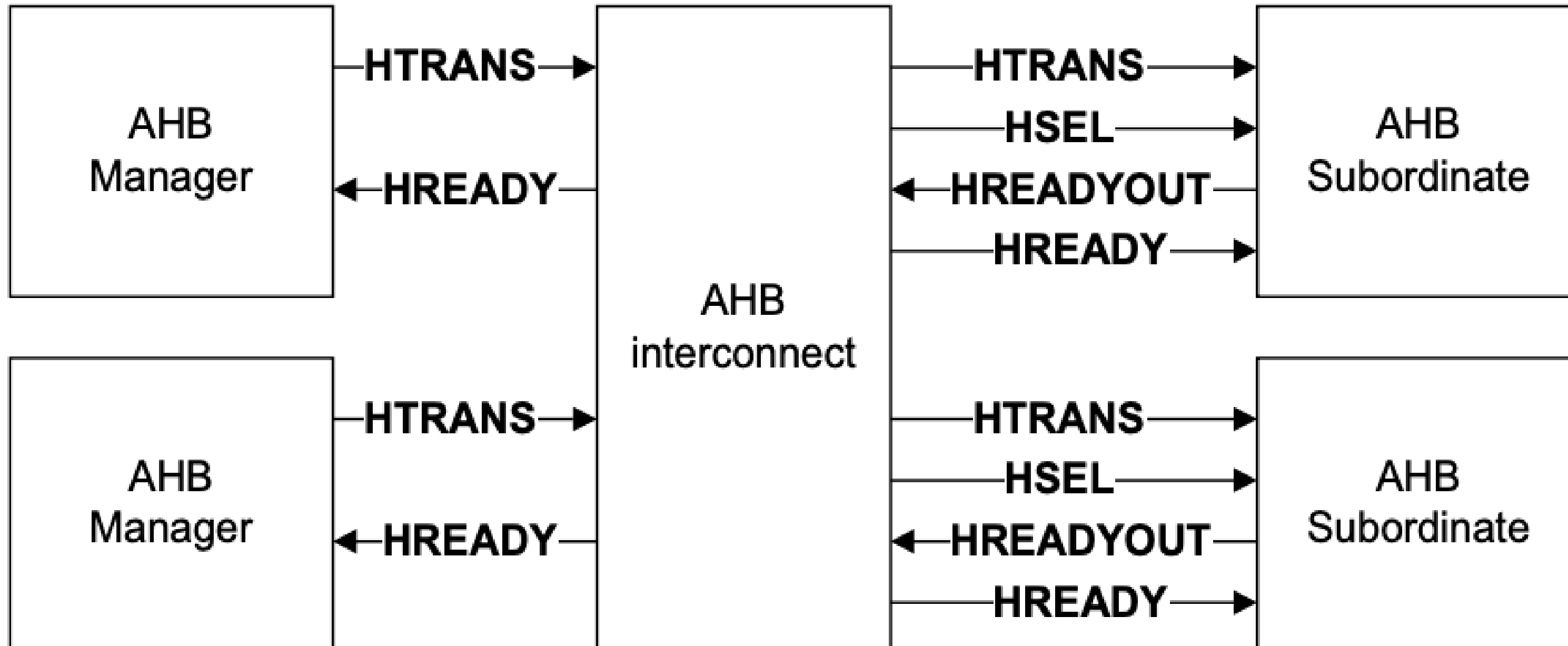
ADDRESS DECODING - SUBORDINATE SELECT SIGNALS



READ DATA AND RESPONSE MULTIPLEXOR



INTERCONNECT WITH AHB INTERFACES



SUBORDINATE TRANSFER RESPONSES

HRESP	Response	Description
0	OKAY	The transfer has either completed successfully or additional cycles are required for the Subordinate to complete the request. The HREADYOUT signal indicates whether the transfer is pending or complete.
1	ERROR	An error has occurred during the transfer. The error condition must be signaled to the Manager so that it is aware the transfer has been unsuccessful. A two-cycle response is required for an error condition with HREADYOUT being asserted in the second cycle.

HRESP	HREADYOUT	
	0	1
0	Transfer pending	Successful transfer completed
1	ERROR response, first cycle	ERROR response, second cycle

ERROR RESPONSE

